GPQA Prefill Activation Projection

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Setup

This note summarizes the corrected activation-based visualization for the finished 2026-03-21 GPQA prompt-profile pilot. The figure is built from the saved prompt prefill activations, not rollout text.

- Data source: the saved pilot bundle at `/data/scratch/murphy/outputs/cot-loop-detection/gpqa_s09_prob`
- Prompts / rollouts: `48` prompts, `10` rollouts per prompt, `480` rollouts total.
- Activation object: last prefill token at the final transformer layer, read from the saved stacked prefill shard and projected with one shared PCA plane.
- Labels: rollout metadata from `prompt_rollout_archive.jsonl`; prompt-level rates from the prompt-profile JSONLs; GPQA correctness reconstructed from the saved source rows using the recorded `sample_id` and the same choice-shuffle seed (`seed = 0`).

The first two PCA axes explain `15.9%` and `9.7%` of prompt variance.

Main Read

- Correctness is the clearest large-scale pattern in this plane. A coarse left-vs-right slice drops from about `53.7%` rollout correctness (`PC1 < -4`) to `16.7%` (`PC1 > 4`).
- Max-length risk is visible but fragmented. The three prompts with `p_cap >= 0.5` sit in three distinct regions of the plane rather than in one single failure cluster.
- Loop risk is broader and more mixed than cap-hit risk. On this pilot every cap hit is still a loop, but only `34 / 180 = 18.9%` of looped rollouts hit the cap.
- Because repeated rollouts from one prompt share one prefill coordinate, the plot should be read as a prompt-propensity view, not as a within-prompt branching view.

Figures

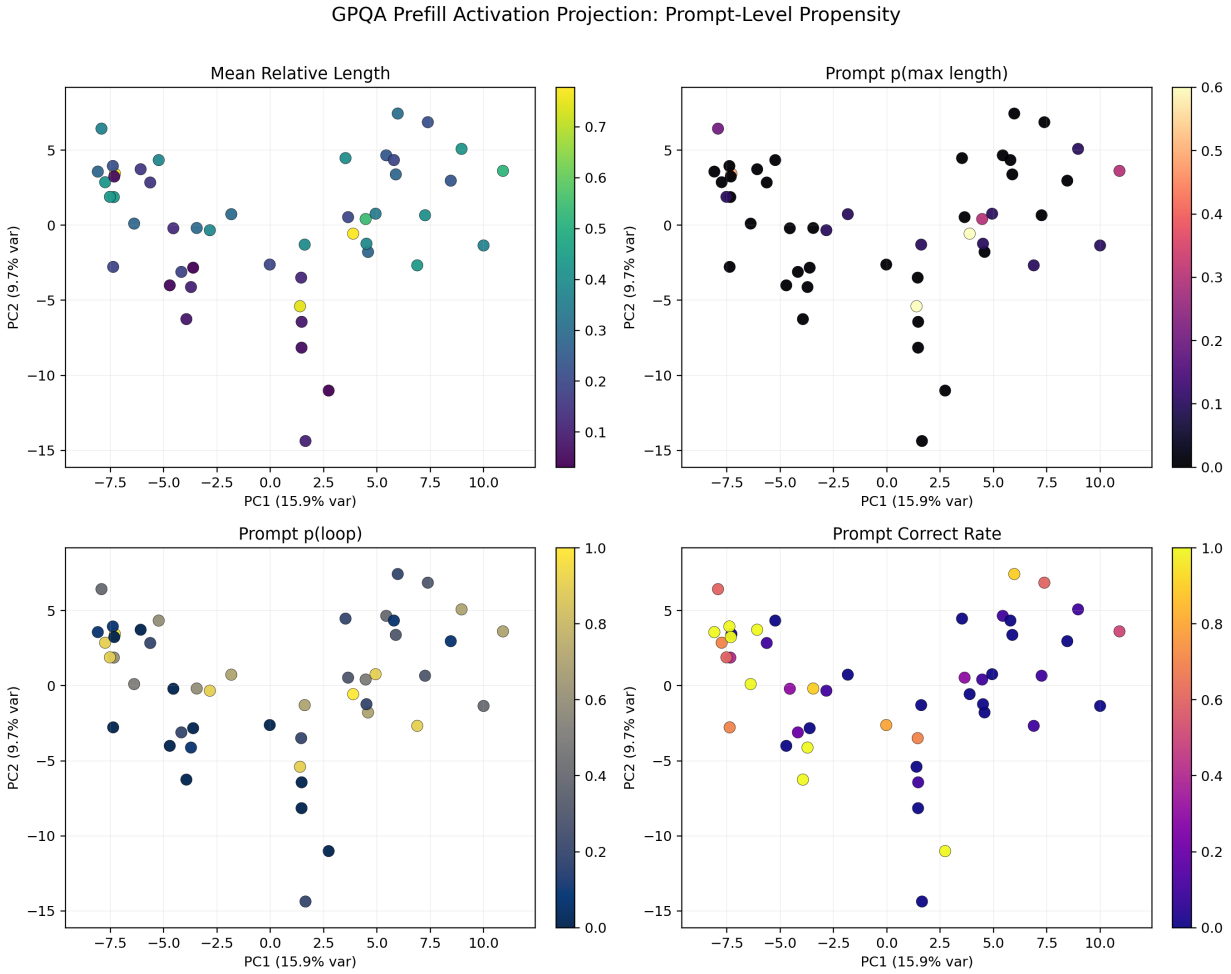

Figure 1: Prompt-level propensity on the shared prefill PCA plane.

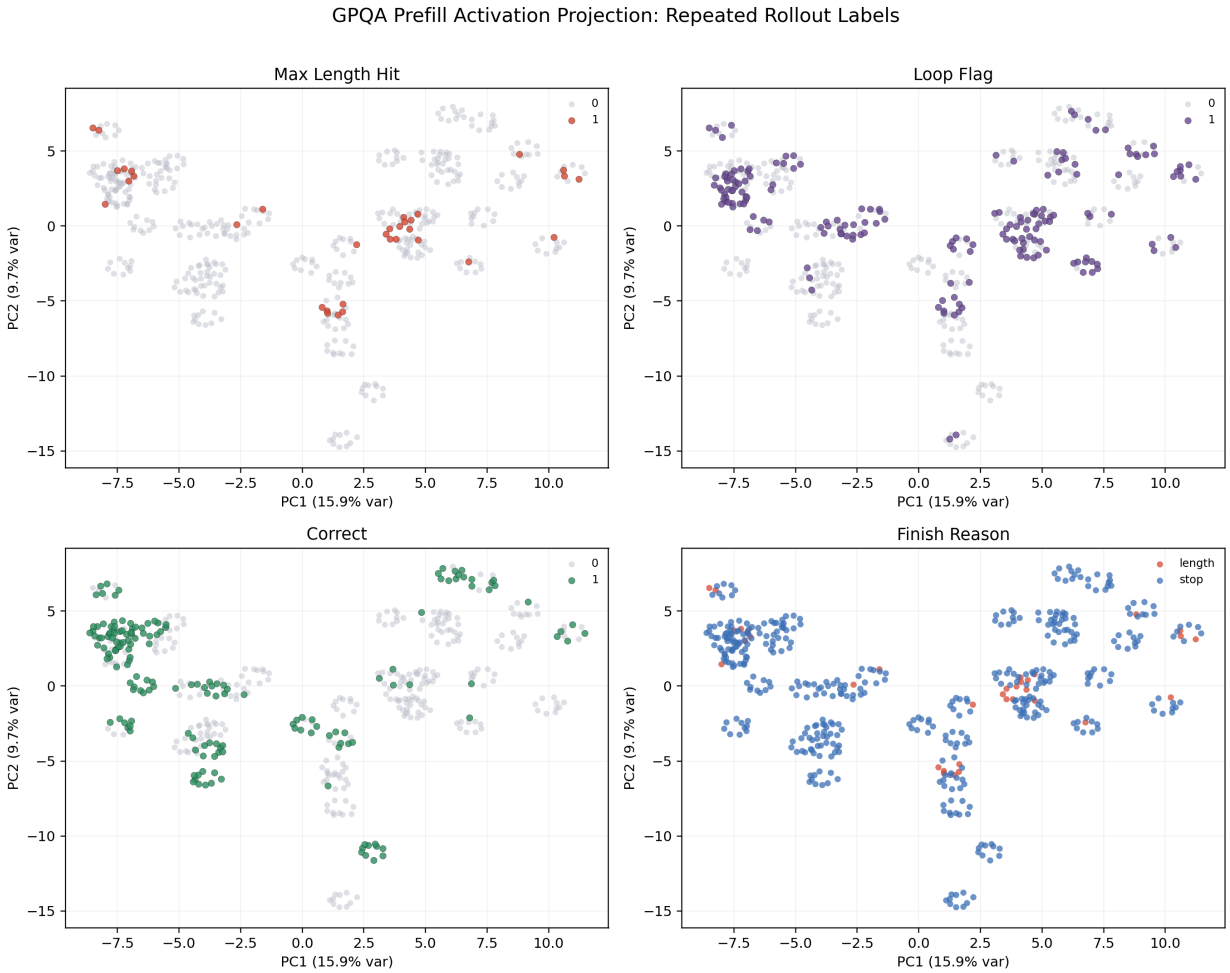

Figure 2: Repeated-rollout labels duplicated on the same prompt prefill coordinates with light deterministic jitter.